

Stem-and-Leaf Diagram

Stem-and-Leaf Display

- A stem-and-leaf display is a method that is used to rank-order and arrange data into groups.
- Frequency distributions and histograms provide a useful organization and summary of data. However, in a histogram, we lose most of the specific data values.
- Stem-and-leaf displays are another way to represent **quantitative data**. They give more detail because they show the actual data. A stem-and-leaf display allows us to *recover the original data if desired*.

Stem-and-Leaf Display

- In a stem-and-leaf plot each data value is split into a "**stem**"
- and a "**leaf**". The "**leaf**" is usually the last digit of the number
- and the other digits to the left of the "leaf" form the "**stem**".
- To show a **one-digit number** (such as 9) using a stem-and-leaf plot, use a stem of 0 and a leaf of 9.
- To show a **two-digit number** (such as 25), the stem is 2 and the leaf is the 5.
- If the observed value is a **three-digit number** (such as 369), the stem is 36 and the leaf is 9.
- If the observations are accurate to one or more decimal places, such as 23.7, the stem is 23 and the leaf is 7.

Stem-and-Leaf Diagram

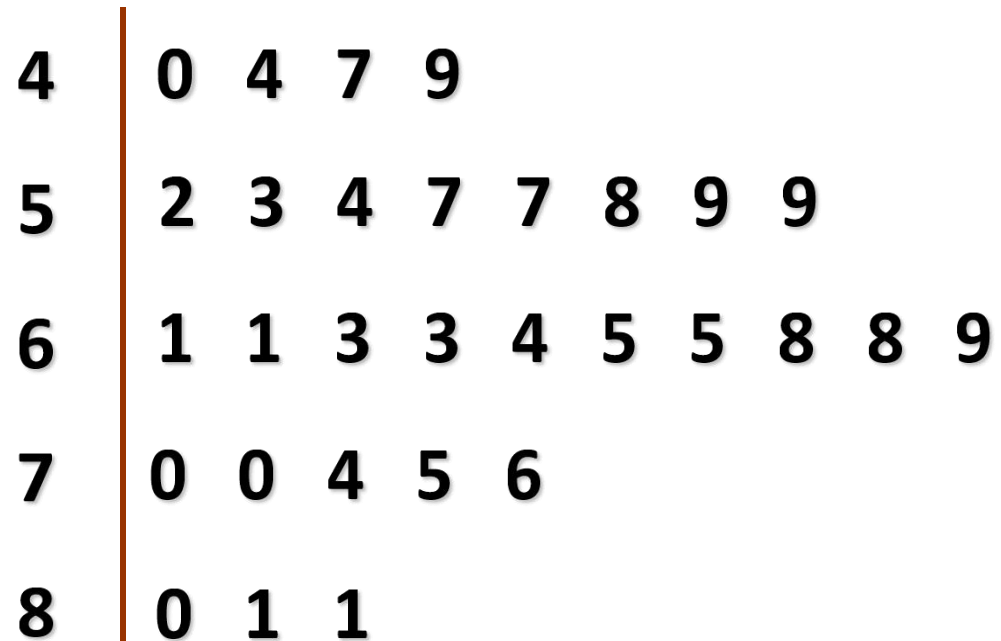
HOW TO MAKE A STEM-AND-LEAF Diagram

1. Divide the digits of each data value into two parts. The **"leaf"** is usually the last digit of the number and the other digits to the left of the **"leaf"** form the **"stem"**.
2. Align all the stems in a vertical column from smallest to largest. Draw a vertical line to the right of all the stems.
3. Place all the leaves with the same stem in the same row as the stem, and arrange the leaves in increasing order.
4. Use a label to indicate the magnitude of the numbers in the display. We include the decimal position in the label rather than with the stems or leaves.

Stem and Leaf Diagrams

Here are the marks gained by 30 students in an examination:

63 58 61 52 59 65 69 75 70 54 57 63 76 81 64
68 59 40 65 74 80 44 47 53 70 81 68 49 57 61



Stem and Leaf Diagrams

How to Draw One:

1. Put the **first digits** of each piece of data in **numerical order** down the left-hand side
2. Go through each piece of data in turn and put the remaining digits in the proper row
3. Re-draw the diagram putting the pieces of data in the **right order**
4. Add a **key**

Stem and Leaf Diagrams

Here are the marks gained by 30 students in an examination:

63 58 61 52 59 65 69 75 70 54 57 63 76 81 64

68 59 40 65 74 80 44 47 53 70 81 68 49 57 61

Write the tens figures in the left hand column of a diagram.

These are the 'STEMS'

4

5

6

7

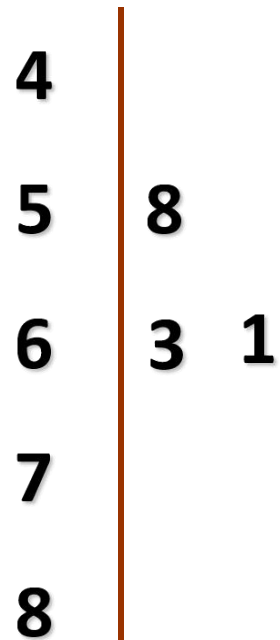
8

Stem and Leaf Diagrams

Here are the marks gained by 30 students in an examination:

~~64~~ ~~59~~ ~~61~~ 52 59 65 69 75 70 54 57 63 76 81 64
68 59 40 65 74 80 44 47 53 70 81 68 49 57 61

Go through the marks in turn and put in the units figures of each mark in the proper row. These are the 'LEAVES'

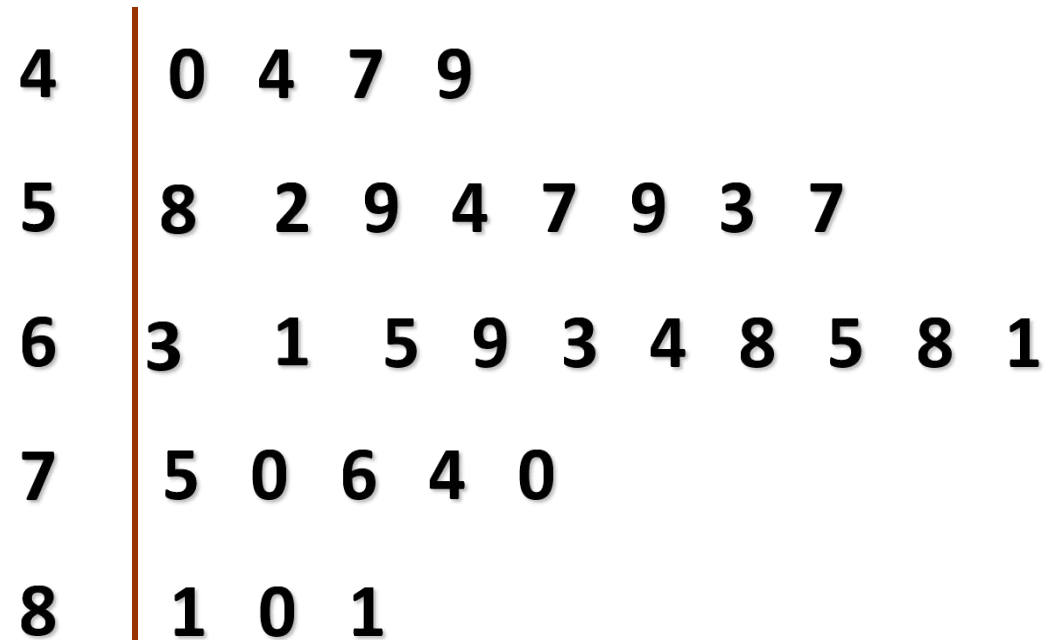


Stem and Leaf Diagrams

Here are the marks gained by 30 students in an examination:

63 58 61 52 59 65 69 75 70 54 57 63 76 81 64
68 59 40 65 74 80 44 47 53 70 81 68 49 57 61

When all the marks are entered the diagram will look like this:



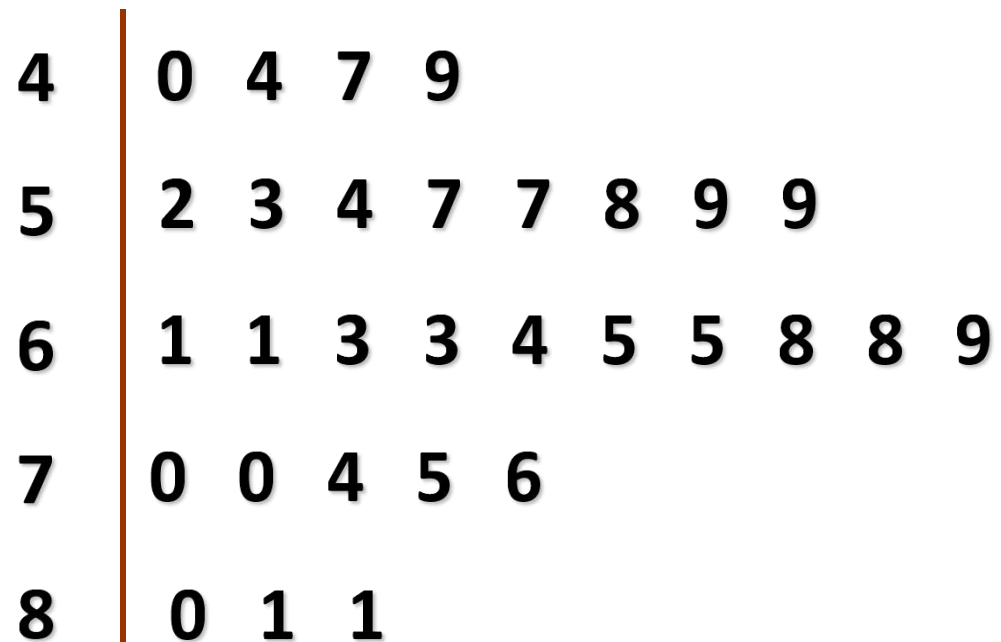
Stem and Leaf Diagrams

Here are the marks gained by 30 students in an examination:

63 58 61 52 59 65 69 75 70 54 57 63 76 81 64

68 59 40 65 74 80 44 47 53 70 81 68 49 57 61

Rewrite the diagram so that the units figures in each row are in order:



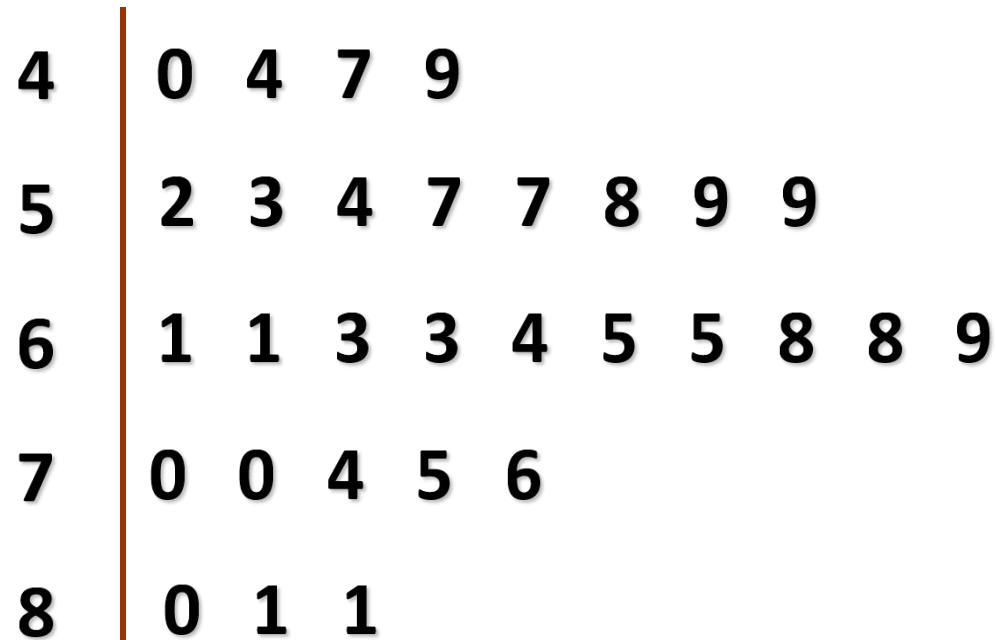
Stem and Leaf Diagrams

Here are the marks gained by 30 students in an examination:

63 58 61 52 59 65 69 75 70 54 57 63 76 81 64
68 59 40 65 74 80 44 47 53 70 81 68 49 57 61

Add a KEY:

5|2 = 52



Stem and Leaf Diagrams

Remember:

- Always put in a **Key**
- Always put your data in **Order**

Median:

- to work out the median, you must find the middle value
- if there are two middle values, you need the average

Range:

- to work out the Range, subtract the smallest number from the biggest

Here are the scores for a freshman basketball team.
Make a stem and leaf plot.

Points Earned in Games During Basketball Season						
27	16	34	29	48	12	33
20	18	42	51	27	32	41

Stem and Leaf plot

Points Earned in Games During Basketball Season						
27	16	34	29	48	12	33
20	18	42	51	27	32	41

Points Earned				
Stem	Leaves			
1	2	6	8	
2	0	7	7	9
3	2	3	4	
4	1	2	8	
5	1			

Key: 1 | 2 = 12

- Here is the same data organized into a stem-and-leaf plot.

Don't forget the key!

Test Scores				
75	86	83	91	94
88	84	99	79	86

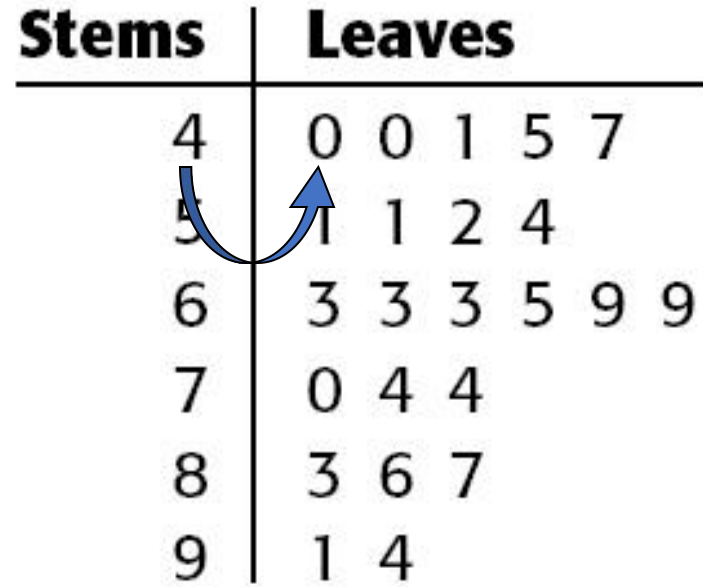
Test Scores

Stems	Leaves
7	5 9
8	3 4 6 6 8
9	1 4 9

Key: 7 | 5 means 75

Using Stem-and-Leaf Plots

Boxes of Girl Scout Cookies Sold by Troop 220



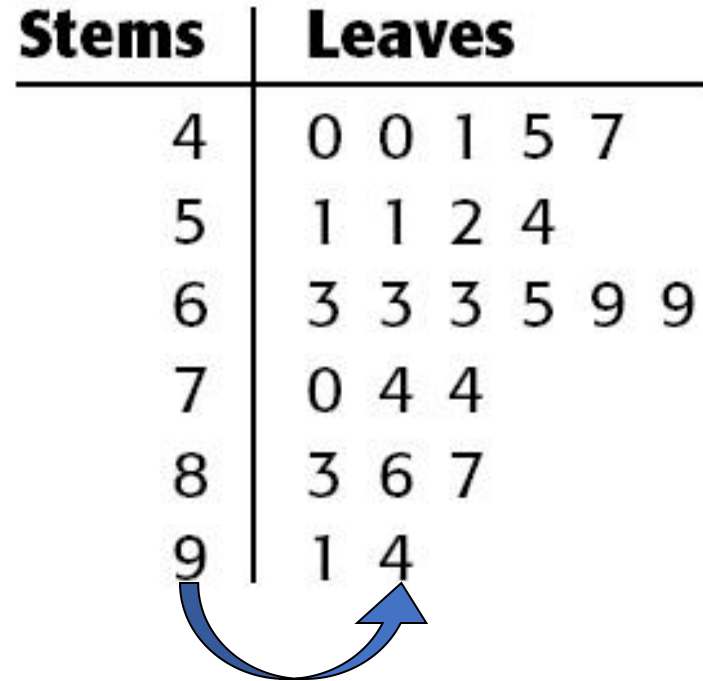
Key: 4|0 means 40

- What was the least number of cookies sold?

40 boxes

Using Stem-and-Leaf Plots

Boxes of Girl Scout Cookies Sold by Troop 220



Key: 4|0 means 40

- What was the largest number of cookies sold?

94 boxes

Using Stem-and-Leaf Plots

Boxes of Girl Scout Cookies Sold by Troop 220

Stems	Leaves
4	0 0 1 5 7
5	1 1 2 4
6	3 3 3 5 9 9
7	0 4 4
8	3 6 7
9	1 4

Key: 4|0 means 40

- Find the median of the data

Find the middle leaf...

63 boxes

Using Stem-and-Leaf Plots

Boxes of Girl Scout Cookies Sold by Troop 220

Stems	Leaves
4	0 0 1 5 7
5	1 1 2 4
6	3 3 3 5 9 9
7	0 4 4
8	3 6 7
9	1 4

Key: 4|0 means 40

- Find the mode of the data

Look for the number that occurs
most often in a row of leaves

63 boxes

Example – *Stem-and-Leaf Diagram*

- Many airline passengers seem weighted down by their carry-on luggage. Just how much weight are they carrying?
- The carry-on luggage weights in pounds for a random sample of 40 passengers returning from a vacation to Hawaii were recorded (see Table below).

30	27	12	42	35	47	38	36	27	35
22	17	29	3	21	0	38	32	41	33
26	45	18	43	18	32	31	32	19	21
33	31	28	29	51	12	32	18	21	26

Example – *Stem-and-Leaf Diagram*

cont'd

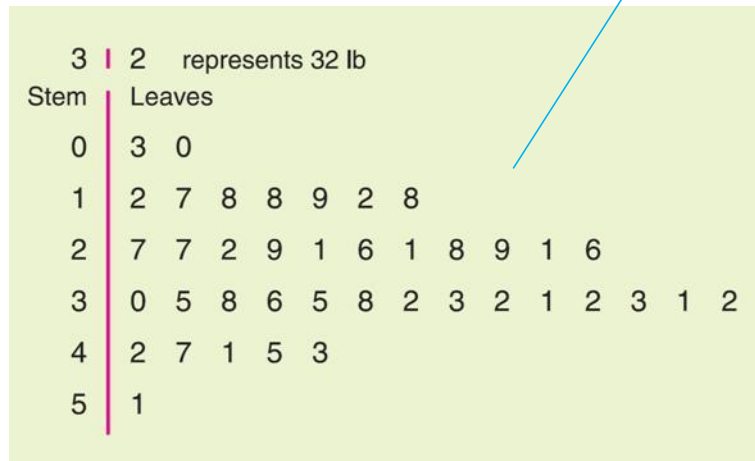
- To make a stem-and-leaf display, we divide each data value into *two* parts.
- The weights in our example consist of two-digit numbers. For a two-digit number, the stem selection is obviously the left digit. In our case, the digit in the tens place will form the *stems*, and the *digit in the units place* will form the *leaf*.
- For example, for the weight 12, the stem is 1 and the leaf is 2. Since the data range from 0 to 51, the stems range from 0 to 5.

Example – *Stem-and-Leaf Diagram*

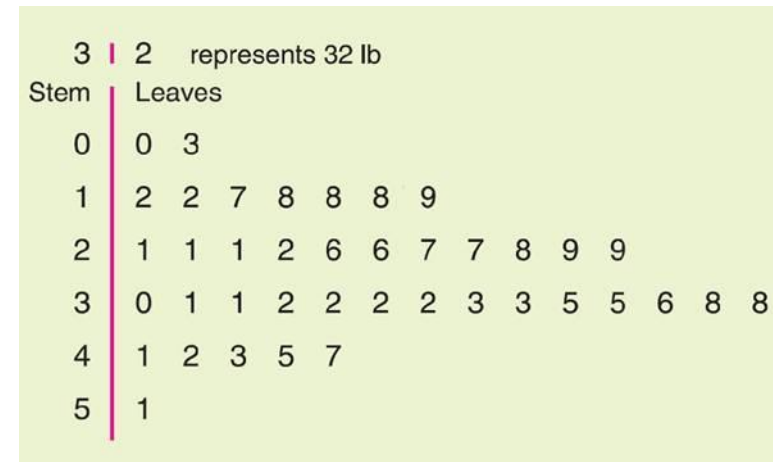
cont'd

- In the stem-and-leaf display, we list each possible stem once on the left and all its leaves in the same row on the right, as in Figure 2-15(a). Then we order the leaves as shown in Figure 2-15(b).

Note: Do not separate the leaves by commas.



(a) Leaves Not Ordered



(b) Final Display with Leaves Ordered

Stem-and-Leaf Displays of Airline Carry-On Luggage Weights

Figure 2-15

Example – *Stem-and-Leaf Diagram*

cont'd

- A good Stem-and-Leaf Display shows how the data are spread -- that is, highest number, lowest number, most common number and outliers (a number that lies outside the main group of numbers).
- It is important to note that when there is a repeated number in the data (such as two 12s) then the plot must reflect such (so the plot would look like 1 | 2 2).
- We see that most of the weights were in the 30-lb range, only two were less than 10 lb, and six were over 40 lb.
- As a final step, we need to *indicate the scale*. This is usually done by indicating the value represented by the stem and one leaf.

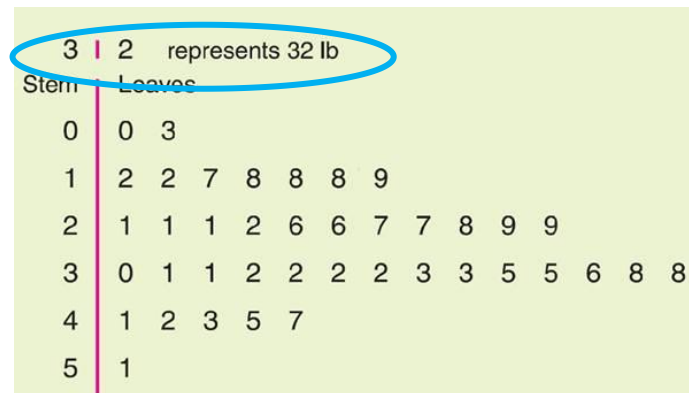
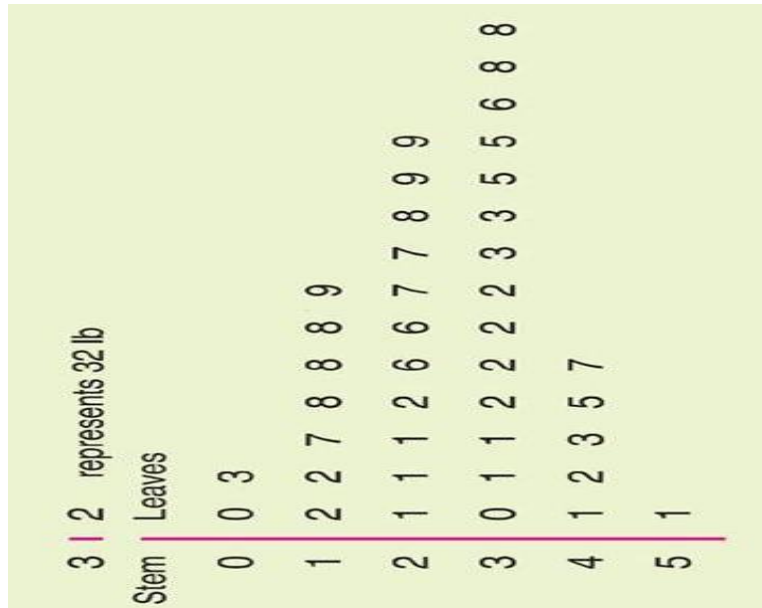


Figure 2-15 (b)

Example – *Stem-and-Leaf Diagram*

- Stem-and-leaf displays show each of the original data values. It's interesting that this plot looks very similar to a histogram, only it gives us the actual data. By looking at the display “**sideways**,” you can see the distribution shape of the data.



Look at the distribution in this example: we say that the distribution is kind of skewed to the left.

Stem-and-Leaf Display

There are some limitations to stem-and-leaf displays.

In particular, we're limited to small data sets - can you imagine the leaves if we had 1,000 test scores?

Also, the range in the data needs to be fairly small.

Stem and Plot with few stems

- **Splitting Stems** – (too few stems or classes) Split stems to double the number of stems when all the leaves would otherwise fall on just a few stems.
 - Each stem appears twice.
 - Leaves 0-4 go on the 1st stem and leaves 5-9 go on the 2nd stem.

Stem and Plot with few stems

Example: data

120,121,121,123,124,124,125,125,125,126,126,128,129,130,132,132,133,134,134,132,
135,137,138,138,138,139

StemPlot

12		0 1 13445556689
13		0223444578889

StemPlot (splitting stems)

12		0 1 1344
12		5556689
13		0223444
13		578889

Splitting the stems

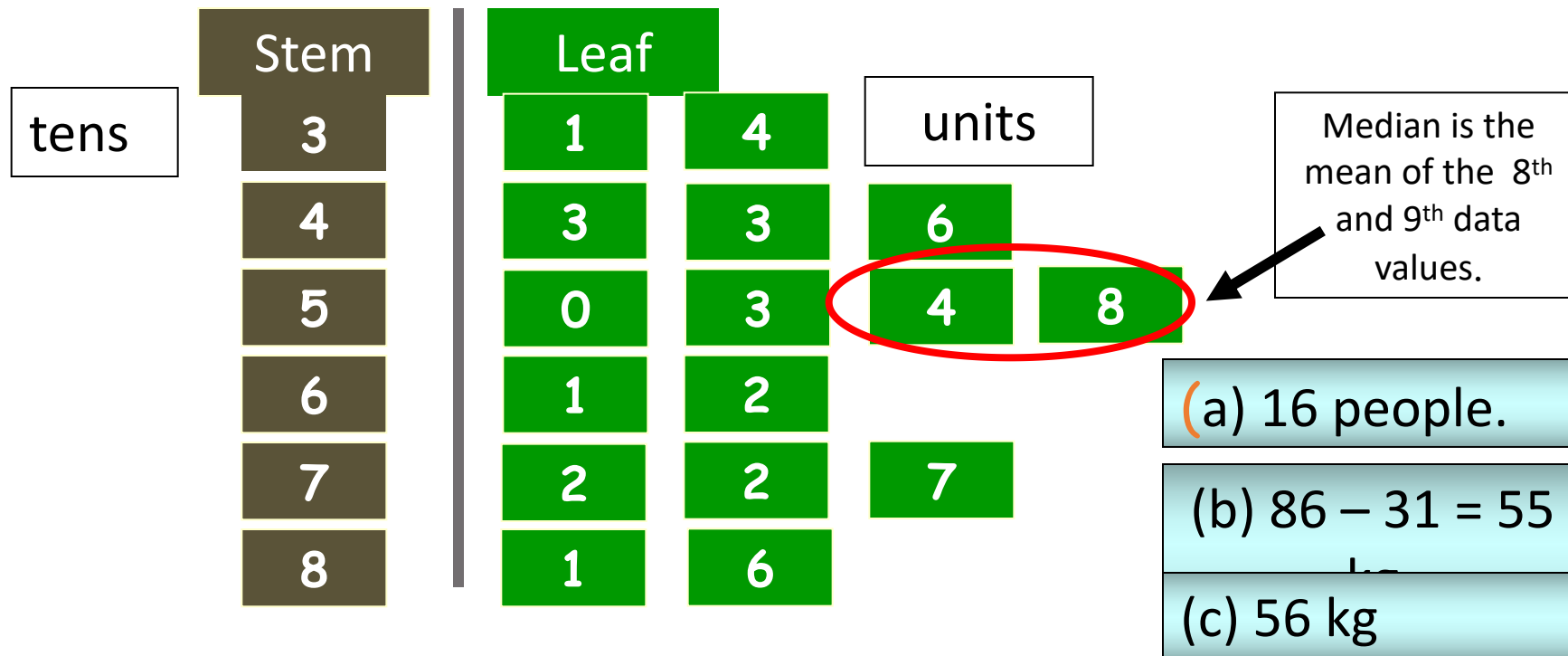
- The stem can be further divided if necessary. The distinction of the stems is made by characters taken from the English language (Tukey, 1977):
- 0 and 1 is *
- 2 and 3 is T,
- 4 and 5 is F
- 6 and 7 is S
- 8 and 9 is .

*	0	1
T	2	3
F	4	5
S	6	7
.	8	9

Stem and Leaf

The stem & leaf diagram below shows the masses in kg of some people in a lift.

- (a) How many people were weighed?
- (b) What is the range of the masses?
- (c) Find the median mass.



Practice problem #1:

One sportswriter listed the winning scores of the conference championship games over the last 35 years. The scores for those games follow below.

132	118	124	109	104	101	125	83	99	131
98	125	97	106	112	92	120	103	111	117
135	143	112	112	116	106	117	119	110	105
128	112	126	105	102					

i) Make a Stem-and-leaf display.

ii) Looking at the distribution, would you say that it is fairly symmetrical?

Practice problem #1:

i) Make a Stem-and-leaf display.

1). We break the digits of each data value into *two* parts. Use the leftmost two digits as *stem*, and the single rightmost digit as *leaf*.

2). Since the data range from 83 to 181, the stems range from 8 to 18. Align all the stems in a vertical column from smallest to largest. Draw a vertical line to the right of all the stems. Write each leaf in the row to the right of its stem.

Stem	Leaves
8	3
9	9 8 7 2
10	9 4 1 6 3 6 5 5 2
11	8 2 1 7 2 2 6 7 9 0 2
12	4 5 5 0 8 6
13	2 1 5
14	3

Practice problem #1:

3). Rearrange the leaves in increasing order. And indicate the scale.

8		3 represents 83 winning points
Stem		Leaves
8		3
9		2 7 8 9
10		1 2 3 4 5 5 6 6 9
11		0 1 2 2 2 2 6 7 7 8 9
12		0 4 5 5 6 8
13		1 2 5
14		3

ii) Looking at the distribution, would you say that it is fairly symmetrical?

Yes. Notice that stem 11 has the most data.

Practice problem

The weights (to the nearest tenth of a kilogram) of 30 students were measured and recorded as follows:

59.2 61.5 62.3 61.4 60.9 59.8 60.5 59.0 61.1 60.7 61.6
56.3 61.9 60.4 62.2 58.9 59.0 61.2 62.1 61.4 58.4 60.8
60.2 62.7 60.0 59.3 61.9 61.7 58.4 59.9

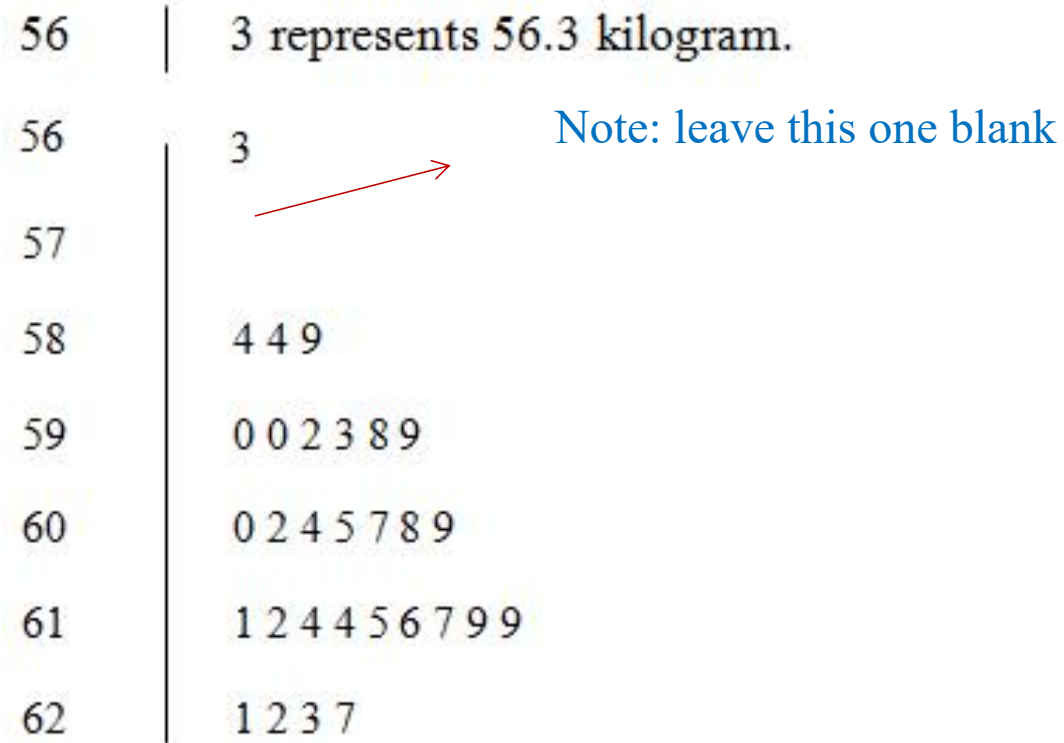
Draw a stem-and-leaf display for the data. Briefly comment on what the analysis shows.

In this case, the stems will be the whole number values and the leaves will be the decimal values.

The data range from 56.3 to 62.7, so the stems should start at 56 and finish at 62.

Practice problem

Weights of 30 students



We say that the distribution is skewed to the left. This stem and leaf display reveals that the group with the highest number of observations recorded is the 61.0 to 61.9 group.

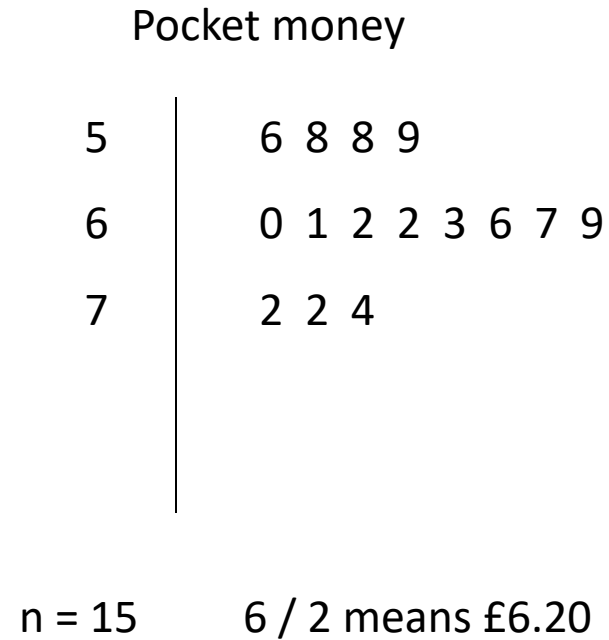
A group of students were asked how much pocket money they get. The results were.

£6.60 £6.90 £6.20 £5.80 £7.40 £5.60 £6.70 £7.20

£6.10 £6.20 £5.90 £6.00 £7.20 £5.80 £6.30

Draw a Stem and Leaf diagram to show this

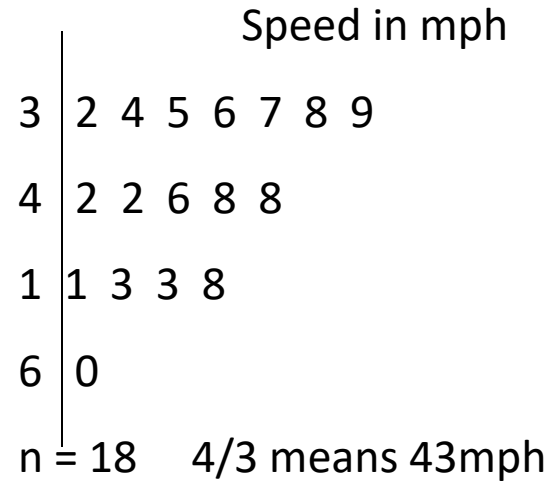
Note this gives us the same diagram as the previous example. This shows the importance of the title and key.



Draw Stem and Leaf Diagrams for the following sets of data

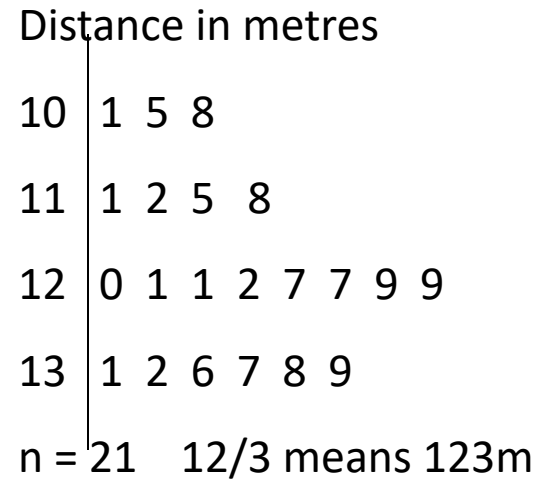
The speed of cars, in mph, passing police speed check

35 46 32 48 53 58 60 39 36 48 34 51 42
37 53 42 51 38



John took a note of how far, in metres, he was able to hit a golf ball

108 115 132 121 101 122 138 127 105
112 118 127 129 120 139 136 131 121
111 129 137



Stem and Leaf diagrams can be used to compare 2 sets of data.

To do this we draw a back to back stem and leaf diagram.

Maths marks for class 1A

8
8 5 3
9 4 4 5 2 0
6 4 3 1 1 0
7 3 3

n = 19

0

1

2

3

4

3/2 means a mark of 32

Maths marks for class 1B

6 8 9
1 2 2 2 5 5 8
0 0 1 1 6 7
3 7 7
1

n = 20

How many students in 1A had a mark of 31?

2

How many students in 1B had a mark of 12?

3

Which class appears to be best at Maths?

1A